

PCT/PCM 10: Two Drawings, Every Network

4. Interpret the Figure

In a residual block, the identity path carries the original input x directly to the addition operation at the output, where it is added to $F(x)$. In an encoder-decoder, skip connections carry spatial details and fine-grained features from encoder layers to decoder layers, helping recover information that the bottleneck may lose.

5. Parameter Calculation

The decoder receives 64 channels from below and concatenates a 64-channel skip connection, giving $64 + 64 = 128$ input channels.

For a 3×3 convolution producing 64 output channels, with one bias per output channel:

$$\text{Parameters} = (3 \times 3 \times 128 + 1) \times 64$$

$$= (1152 + 1) \times 64$$

$$= 1153 \times 64$$

$$= 73,792 \text{ parameters.}$$

6. Prediction

If $F(x)$ learns to output zero, the residual block computes $y = F(x) + x = 0 + x = x$, so it behaves like an identity mapping. This helps very deep networks because a block can preserve its input when additional transformations are not useful, making optimization and gradient flow easier.